

Movement Path Estimator

A Hybrid Optical Flow Approach

Solving the Enz Technik Challenge for Autonomous Sewer Inspection.

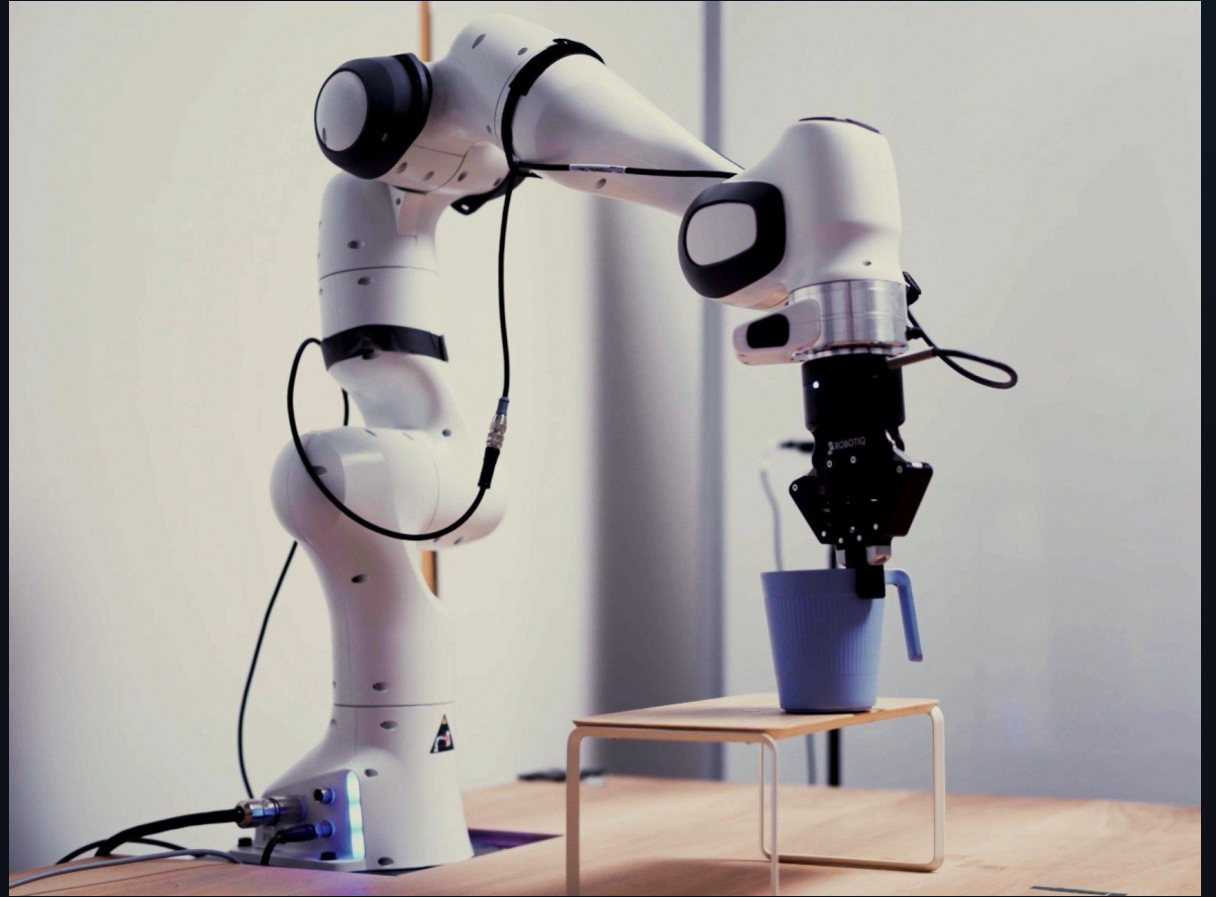
Robust, hardware-accelerated computer vision for dynamic
environments.

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Navigating the **Physical World**

- ▶ **The Goal:** Output the positional movement path (meters per frame) and identify the exact turning point where the camera reverses direction.
- ▶ **The Environment:** Sewer pipes are highly hostile to standard vision algorithms due to splash water, turbulence, and featureless pipe walls.
- ▶ **Our Strategy:** A dual-stream architecture combining the robustness of Deep Learning with the precision of Classical Computer Vision.



The Dual-Stream Ensemble



Stream 1: RAFT

Deep learning model running in batches on the GPU to compute dense optical flow, capturing overall pixel movement robustly.



Stream 2: Lucas-Kanade

Classical algorithm running on the CPU to compute sparse optical flow, tracking specific, high-contrast features precisely.

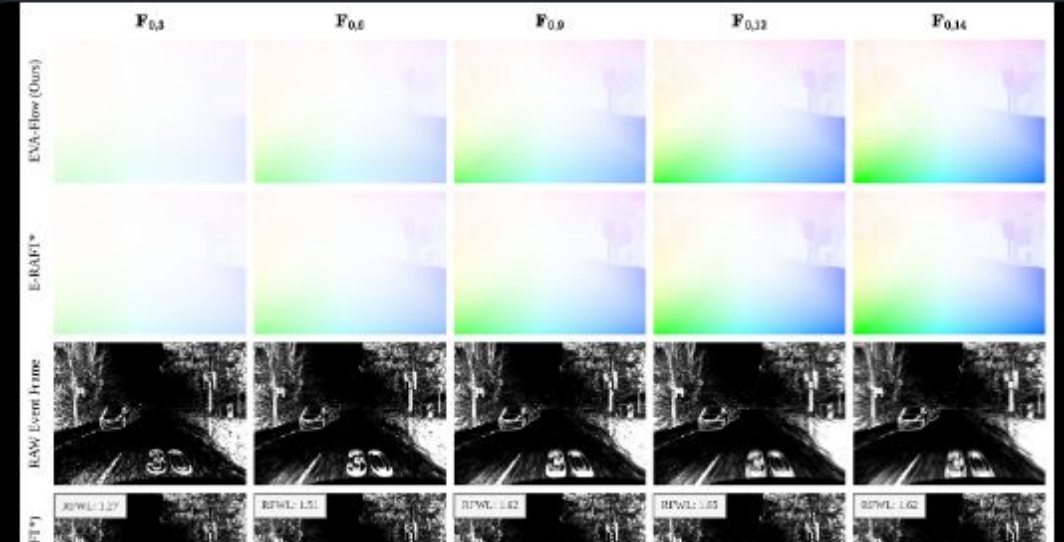


Data Fusion

A sophisticated signal-processing pipeline that normalizes, filters, and combines both streams to determine the final mechanical path.

Stream 1 – Deep Learning with **RAFT**

- ▶ **Why RAFT?** State-of-the-art accuracy for estimating motion. We use *raft_small* to prioritize processing speed.
- ▶ **GPU Batching:** Frames are passed to the model in dynamic batches (size = 32) to maximize GPU utilization and prevent VRAM exhaustion.
- ▶ **Radial Velocity Extraction:** Standard x/y panning is irrelevant in a straight pipe. We isolate forward/backward motion radially.



Stream 2 – Classical CV with **Lucas-Kanade**

Sparse Feature Tracking

To complement the neural network, we implement the Lucas-Kanade (LK) algorithm via OpenCV. The mechanism identifies up to 200 "good features to track" (corners, edges, anomalies) in one frame and precisely calculates their translation in the subsequent frame.

Hardware Efficiency

While RAFT excels at understanding global scene movement, LK is highly precise at tracking distinct physical points (like pipe joints). By executing LK exclusively on the CPU, we achieve true parallel processing, freeing up the GPU entirely for the RAFT batches.

Engineering for the **Physical Reality**



Water/Turbulence Filter

The bottom 40% of the frame is entirely masked out on the GPU to ignore flowing water and spray that confuses flow estimators.



Center Void & Edge Distortion

The direct center and extreme outer edges are ignored due to severe lens distortion and lack of reliable depth perspective.



Fast I/O Tensor Stacking

To prevent disk-read bottlenecks, all frames are pre-loaded, resized, normalized, and stacked into a single giant tensor in RAM.

Translating **Pixels** to **Meters**

Asymmetric Thresholding

The physical crawler behaves differently going forward than backward. We use an asymmetric threshold, making the system 40% more sensitive to detecting backward drag than forward motion.

State Detection

Once the combined velocity stream is generated, frames are classified into distinct mechanical states: Forward (1), Backward (-1), or Stationary (0).

The Return Anchor

We know the crawler must return to the start. We implemented a "Return Trip Anchor" that algorithmically forces the final path error down to 0 meters, mathematically smoothing out cumulative drift.



Efficiency, Accuracy, and Novelty

Our solution directly addresses the hackathon's core criteria: Ensemble median filtering eliminates drift, pre-loading tensors and CPU/GPU split processing guarantees efficiency, and the Annulus Mask bridges the gap between raw data and physical reality.